

Tableau

Objectives

The notebook builds on previous weeks by formalizing the model into a production-ready pipeline and extending the analysis with unsupervised learning (FP-Growth association rule mining) and professional interactive visualization using Plotly. These additions directly addresses the Chapter 9 topics on unsupervised learning and external tool integration.

Target Variable: $PM_{2.5}$

Features used: CO, NO_2 , SO_2

Objective	Chapter 9 tie-In	Delegated to
Validate model stability using 5-Fold Cross-Validation	Algorithm Metrics	Polly and Erick

Tasks Completed

1. Implemented 5-Fold Cross-Validation: Used 'cross_val_score' on the 'model pipeline' to assess performance across five non-overlapping folds of the training data.
2. Calculated Aggregate Metrics: Derived the Mean R^2 and Standard Deviation to quantify the average expected performance and the model's variance.
3. Stability Check: Integrated a conditional logic check in order to verify if the calculated standard deviation was below the threshold of 0.01.

Tasks to be Completed

All Tasks have been completed.

Results and Analysis

```
... Running 5-Fold Cross-Validation on training data...

— Cross-Validation Results ( $R^2$ ) —
Fold 1: 0.3541
Fold 2: 0.3587
Fold 3: 0.3595
Fold 4: 0.3550
Fold 5: 0.3541

Mean : 0.3563
Std  : 0.0023

→ Low variance across folds: model is stable and generalizes well.
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The low standard deviation of 0.0023 across the 5 folds indicates that the Random Forest Regressor model exhibits strong stability and generalizes well across different subsets of the training data. This consistent performance builds confidence in the model's reliability, suggesting that it is not overly sensitive to the specific data split and can be expected to perform similarly on unseen data. The low mean R^2 shows that while the model is not yet capable of accounting for all the variance, its reliability displays a potential for further growth provided a better training environment and fed with the necessary dataset.

Challenges and Solutions

Challenge: RAM Constraints during Model Training

Solution: Lowering the folds required for cross-validation. Though it does generate a less optimal data.

Conclusion

The objective of validating model stability using 5-Fold Cross-Validation was successfully met. The results, characterized by a consistent mean R^2 and a very low standard deviation, confirm that the model is robust and its performance is stable across various non-overlapping folds. This exercise reinforced the importance of cross-validation as a technique to obtain a more reliable estimate of model performance and to detect potential issues like overfitting or high variance.